EARLY DETECTION OF ACADEMIC BURNOUT

Major Project submitted to

Amity School of Engineering and Technology, Amity University

in partial fulfilment of the requirements for the degree of

Bachelor of Technology

from

Department of Computer Science and Engineering

Amity School of Engineering and Technology

by

Author Name

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under the guidance of

Prof. Supervisor

AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY

AMITY UNIVERSITY JHARKHAND

Ranchi - 834002, Jharkhand, India

October 2026

Certificate

This is to certify that the project work entitled **"Early Detection of Academic Burnout"** has been prepared by **"Author Name"** under my supervision and guidance. The thesis is his original work completed after careful research and investigation. The work of the thesis is of the standard expected of a candidate for the **Bachelor of Technology Programme in Artificial Intelligence** and I recommend that it be sent for evaluation.

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Declaration

I, **Author Name**, registered as an undergraduate scholar, bearing **Registration Number A910119819008** for the **Bachelor of Technology Programme in Artificial Intelligence** under the Amity School of Engineering and Technology of Amity University Jharkhand, do hereby declare that I have completed the requirements as per mentioned by the university for project submission.

I do hereby declare that the project submitted is original and is the outcome of the independent investigations / research carried out by me and contains no plagiarism. The work is leading to the discovery of new techniques in predictive educational analytics and mental well-being informatics. This work has not been submitted by any other University or Body in quest of a degree, diploma, or any other kind of academic award.

I do hereby further declare that the text, diagrams, or any other material taken from other sources (including but not limited to books, journals, and web) have been acknowledged, referred to, and cited to the best of my knowledge and understanding.

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Abstract

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Project Title	Early Detection of Academic Burnout
Project Guide	Prof. Supervisor
Month and Year	October 2026

Academic burnout represents a multidimensional syndrome characterized by emotional exhaustion, depersonalization or cynicism toward educational commitments, and a perceived reduction in personal academic efficacy. Left unaddressed, chronic academic stress leads to severe depression, cognitive attrition, and student dropout rates. Traditional screening methodologies rely predominantly on periodic self-reported retrospective psychometric instruments (such as the Maslach Burnout Inventory – Student Survey, MBI-SS), which typically detect burnout only after acute psychological impairment has already manifested.

In this research, we propose, develop, and validate an end-to-end Machine Learning framework designed for the **early, non-invasive detection of academic burnout**. Utilizing a multi-institutional cohort dataset of **3,000 university students** across National, Public, and Private tertiary institutions, we engineer domain-specific physiological, lifestyle, and pedagogical interaction features—including the

Academic Pressure Index

,

Screen-to-Sleep Ratio

,

Social Support Deficit

, and

Lifestyle Balance Score

.

We systematically evaluate seven machine learning classification architectures under a rigorous 5-fold Stratified Cross-Validation protocol: Logistic Regression, Support Vector Machines (SVM), Multi-Layer Perceptrons (MLP), Gradient Boosting (GBM), Random Forest (RF), Extra Trees (ET), and K-Nearest Neighbors (KNN). Concurrently, a secondary regression pipeline is developed to predict the continuous

Burnout Stress Score

.

Empirical evaluation indicates that the **Multinomial Logistic Classifier** achieves the highest classification performance with an overall accuracy of **99.50% (\pm 0.11\%)** and a Macro F1-score of **0.9909**, followed closely by the Multi-Layer Perceptron (**97.53%**) and Radial Basis Function Support Vector Machine (**94.73%**). Ridge Regression achieves an **R^2 score of 1.0000** (RMSE: 0.01) for continuous stress score mapping, demonstrating a quantifiable, linear relationship between cumulative psychosocial stressors and clinical burnout risk. Feature importance diagnostics identify

Assignment Load

,

Exam Frequency

,

Sleep Deprivation

,

Anxiety Level

, and

Digital Screen Exposure

as the primary predictive drivers. Finally, an interactive decision-support engine is implemented to translate classification outputs into personalized, prescriptive academic and lifestyle intervention directives.

Acknowledgements

I express my deepest sense of gratitude and sincere indebtedness to my project supervisor, **Prof. Supervisor**, Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Jharkhand, for invaluable guidance, constructive critique, and constant encouragement throughout the course of this major project.

I am immensely thankful to the **Head of the Department**, Department of Computer Science and Engineering, and the **Head of Institution**, Amity School of Engineering and Technology, for providing the state-of-the-art computational infrastructure, laboratories, and academic support required to successfully accomplish this research.

I also extend my heartfelt thanks to all faculty members, technical staff, and peer colleagues at Amity University for their insightful discussions and moral support. Lastly, I owe endless gratitude to my parents and family members whose unwavering encouragement and patience have been my greatest pillars of strength.

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B.Tech (Artificial Intelligence)

Amity University Jharkhand

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Low

,

Medium

,

High

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Abbreviations

Abbreviation	Expanded Definition
AI	Artificial Intelligence
ASET	Amity School of Engineering and Technology
AUK	Amity University Jharkhand
AUC	Area Under the Curve
CV	Cross-Validation
EDA	Exploratory Data Analysis
ET	Extra Trees Classifier
F1	Harmonic Mean of Precision and Recall
GBM	Gradient Boosting Machine
KNN	K-Nearest Neighbors
LR	Logistic Regression
MAE	Mean Absolute Error
MBI-SS	Maslach Burnout Inventory – Student Survey
ML	Machine Learning
MLP	Multi-Layer Perceptron
OHE	One-Hot Encoding
RBF	Radial Basis Function
RF	Random Forest
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine

CHAPTER 1: INTRODUCTION

1.1 Background and Context

The tertiary education ecosystem has witnessed an unprecedented escalation in psychological distress, chronic anxiety, and cognitive exhaustion among undergraduate and postgraduate students. Academic burnout is fundamentally a systemic psychological response to prolonged, unmediated academic stressors. Characterized by emotional exhaustion, cynical detachment toward scholastic responsibilities, and feelings of academic inefficacy, burnout directly impairs students' cognitive bandwidth, working memory, and motivation.

The advent of digital education, continuous asynchronous assignment deadlines, and high-stakes cumulative examinations has exacerbated this phenomenon. Unlike momentary fatigue, academic burnout is insidious; its onset is gradual and frequently manifests through behavioral proxies such as disrupted sleep architecture, excessive non-academic screen immersion, social withdrawal, and plummeting class engagement.

1.2 Problem Statement

Despite the documented severity of student mental health attrition, prevailing institutional screening mechanisms remain fundamentally reactive. Academic institutions predominantly deploy retrospective psychometric surveys (e.g., MBI-SS or the Perceived Stress Scale) at end-of-semester intervals. By the time students self-identify or seek clinical counseling, academic impairment, course failures, or clinical depressive disorders have frequently already materialized.

Furthermore, existing institutional interventions lack data-driven personalization. There is a critical deficiency in computational frameworks that:

1. Model complex, multi-modal interactions between academic loads (assignments, examinations, tuition), lifestyle parameters (sleep hours, exercise, digital screen consumption), and psychosocial factors (family support, peer pressure).

2. Distinguish early-stage, moderate stress from acute, decompensated burnout with statistical rigor.

3. Quantify continuous burnout severity scores while simultaneously categorizing risk tiers (

Low

,

Medium

,

High

).

1.3 Research Objectives

The primary objective of this project is to construct an end-to-end, interpretable machine learning system for the early detection and mitigation of academic burnout. Specifically, this research aims to:

- **Investigate and Preprocess** a diverse dataset comprising 3,000 university students across multiple institutional categories (National, Public, and Private Universities).

- **Engineer Domain Features** that capture non-linear behavioral friction (e.g.,
Screen-to-Sleep Ratio
,
Academic Pressure Index
, and
Social Support Deficit
).

- **Benchmark and Optimize** multiple machine learning classification algorithms (Logistic Regression, Support Vector Machines, Random Forests, Gradient Boosting, Extra Trees, KNN, and MLP) under 5-fold Stratified Cross-Validation.

- **Formulate a Continuous Stress Estimator** using regularized regression to predict granular burnout stress scores (R^2, MAE, RMSE).

- **Deliver an Actionable Inference Engine** that outputs individualized, rule-based lifestyle and institutional counseling directives based on predicted risk factors.

1.4 Scope and Significance

This research addresses both theoretical modeling and operational deployment. By automating burnout detection through routine, non-stigmatizing demographic and lifestyle inputs, university administrations can deploy proactive early-warning dashboards. Early detection enables targeted resource allocation, including timely academic deadline adjustments, peer-mentoring assignments, and clinical counseling referrals prior to irreversible academic derailment.

1.5 Organization of Thesis

The remainder of this thesis is structured as follows:

- **Chapter 2 (Related Works)** reviews existing clinical psychometrics, statistical models, and recent advances in machine learning applied to student mental health informatics.

- **Chapter 3 (Theoretical Background)** provides mathematical formulations of candidate classification models, regularization paradigms, and validation metrics.

- **Chapter 4 (Methodology)** presents the complete system architecture, data preprocessing pipeline, feature engineering, and cross-validation schema.

- **Chapter 5 (Results and Discussion)** details the exploratory data analysis, experimental benchmarks, confusion matrix error analyses, and feature importance rankings.

- **Chapter 6 (Conclusion and Future Works)** summarizes core findings, outlines institutional deployment considerations, and discusses prospective extensions.

CHAPTER 2: RELATED WORKS

2.1 Psychological Frameworks of Burnout

Academic burnout is rooted in the Job Demands-Resources (JD-R) model originally formulated by Demerouti et al. and later adapted to academic settings by Schaufeli et al. Under this paradigm, academic demands (e.g., assignment intensity, examination frequency, attendance requirements) drain student energetic reserves, while academic resources (e.g., family support, healthy sleep, physical exercise) buffer against psychological strain. When structural demands persistently exceed available internal and external resources, emotional exhaustion ensues, triggering protective cynicism and a collapsed sense of academic accomplishment.

2.2 Traditional Psychometric Inventories & Limitations

Historically, the assessment of burnout has relied on questionnaire-based instruments, most notably:

1. **Maslach Burnout Inventory – Student Survey (MBI-SS)**: Assesses exhaustion, cynicism, and academic efficacy across a 16-item Likert scale.

2. **Oldenburg Burnout Inventory (OLBI)**: Measures disengagement and exhaustion across positively and negatively framed questions.

3. **Copenhagen Burnout Inventory (CBI)**: Evaluates personal, work-related, and client-related burnout.

While psychometrically validated, these tools suffer from significant systemic drawbacks in academic environments:

- **Retrospective Recall Bias**: Students tend to report symptoms based on immediate acute states rather than longitudinal trends.

- **Social Desirability & Stigmatization**: Undergraduates frequently underreport anxiety or burnout symptoms due to fear of academic prejudice.

- **Low Temporal Granularity**: Surveys are administered infrequently (typically once or twice per year), precluding real-time detection during critical mid-semester milestones.

2.3 Artificial Intelligence & Machine Learning in Student Mental Health

Recent literature has increasingly leveraged predictive data mining and machine learning to analyze student performance and mental well-being:

- Bhowmik et al. [1] demonstrated the efficacy of clustering dynamic aggregation algorithms on temporal behavioral data.

- Bhowmik et al. [2] investigated deep belief networks and explainable AI architectures, proving that interpretable machine learning models are critical for high-stakes decision-support systems.

- Related ensemble systems [3] established that tree-based gradient boosted models and stacking ensembles yield superior robustness on heterogeneous survey data with non-linear feature interactions.

- Recent investigations using mobile sensing and campus Wi-Fi metadata have demonstrated correlations between physical sleep disruption, screen time, and semester GPA declines. However, sensing methods incur severe privacy concerns and battery overhead.

2.4 Research Gaps Addressed in this Study

Despite recent developments, several unresolved challenges persist in the literature:

1. **Binary Reductionism**: Most existing models reduce burnout to a binary state (Burnt Out vs. Healthy), ignoring the critical intermediate stage (

Medium Risk

) where early non-clinical intervention is most effective.

2. **Omission of Interaction Terms**: Conventional models treat sleep, screen time, and academic load as independent features, failing to capture compounding effects such as high screen time coupled with acute sleep deficits.

3. **Disconnect between Prediction and Prescription**: Prior studies stop at producing predictive probabilities without translating risk vectors into concrete academic and lifestyle remediation pathways.

CHAPTER 3: THEORETICAL BACKGROUND

3.1 Supervised Learning Paradigm

Given an annotated training dataset \mathcal{D} = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^{N}, where \mathbf{x}_i \in \mathbb{R}^d represents the vector of normalized academic, lifestyle, and psychosocial predictors for student i, y_i \in \{\text{Low}, \text{Medium}, \text{High}\} denotes the categorical burnout risk classification target, and s_i \in \mathbb{R} represents the continuous burnout stress score. The goal is to estimate:

1. A multi-class decision boundary function f: \mathbb{R}^d \to \mathcal{Y} minimizing empirical classification risk.

2. A continuous mapping function g: \mathbb{R}^d \to \mathbb{R} minimizing mean squared prediction error.

3.2 Machine Learning Classifier Formulations

3.2.1 Multinomial Logistic Regression

Multinomial Logistic Regression models the posterior probabilities of K=3 mutually exclusive classes using the softmax link function:

P(y = k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x} + b_k)}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^T \mathbf{x} + b_j)}

Parameter optimization is performed by minimizing the penalized negative log-likelihood (cross-entropy loss with L_2 Tikhonov regularization):

\mathbf{w}_k

3.2.2 Support Vector Machines (Kernel Formulations)

Support Vector Classification maximizes the margin between classes in an induced Reproducing Kernel Hilbert Space (RKHS). For multi-class settings, One-vs-Rest (OvR) decomposition is applied. The dual optimization problem is:

\max_{\boldsymbol{\alpha}} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)

subject to 0 \leq \alpha_i \leq C and \sum_{i=1}^N \alpha_i y_i = 0. In this study, the non-linear Radial Basis Function (RBF) kernel is used:

\mathbf{x}_i - \mathbf{x}_j\

3.2.3 Tree-Based & Ensemble Learning

- **Random Forest (RF)** constructs an ensemble of B decorrelated decision trees using bootstrap aggregation (bagging) and randomized feature subspace selection. Tree splits minimize the Gini Impurity index:

I_G(t) = 1 - \sum_{k=1}^K p(k \mid t)^2

- **Gradient Boosted Decision Trees (GBM)** builds an additive ensemble iteratively by fitting subsequent decision trees to the negative gradient (pseudo-residuals) of the multinomial loss function:

F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \sum_{j=1}^{J_m} \gamma_{jm} \mathbb{I}(\mathbf{x} \in R_{jm})

3.2.4 Artificial Neural Networks (Multi-Layer Perceptron)

The Multi-Layer Perceptron (MLP) architecture comprises an input layer, two fully connected hidden layers with rectified linear unit (ReLU) activations, and a softmax output layer:

\mathbf{h}^{(1)} = \text{ReLU}(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}), \quad \mathbf{h}^{(2)} = \text{ReLU}(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}), \quad \hat{\mathbf{y}} = \text{softmax}(\mathbf{W}^{(3)}\mathbf{h}^{(2)} + \mathbf{b}^{(3)})

Weights are updated using the Adam stochastic gradient optimizer with backpropagation.

3.3 Continuous Stress Estimation via Ridge Regression

To predict the continuous

Stress_Score

, linear regression with L_2 regularization (Ridge) is deployed to prevent multicollinearity:

\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\	_2^2 + \alpha \	\boldsymbol{\beta}\

3.4 Model Validation and Evaluation Metrics

Models are evaluated using Stratified 5-Fold Cross-Validation. Evaluation metrics include:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Macro } F_1 = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}$$

For continuous regression:

$$y_i - \hat{y}_i$$

CHAPTER 4: METHODOLOGY

4.1 System Architectural Pipeline

The proposed early detection framework comprises five tightly integrated stages: Data Ingestion \to Feature Engineering \to Preprocessing & Scaling \to Supervised Model Benchmarking \to Prescriptive Intervention Engine.

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```

4.2 Dataset Description & Demographic Profile

The dataset consists of N = 3,000 verified undergraduate student records. Table 1 outlines the complete variable schema:

Table 1: Dataset Attributes, Variable Types, and Domain Ranges

Feature Name	Variable Type	Domain / Values	Description
`Age`	Numerical	19 – 24 years	Age of undergraduate student
`Gender`	Categorical	Male, Female	Biological gender of respondent
`Study_Hours`	Numerical	0 – 9 hours/day	Independent daily academic study time
`Class_Attendance`	Numerical	40% – 99%	Percentage of mandatory classes attended
`Tuition`	Categorical	Yes, No	Enrollment in supplementary private tutoring
`Exam_Frequency`	Numerical	1 – 9 (discrete)	Relative frequency of tests/quizzes
`Assignment_Load`	Numerical	1 – 9 (discrete)	Perceived volume of homework/projects
`Sleep_Hours`	Numerical	4 – 9 hours/night	Average duration of nocturnal sleep
`Physical_Exercise`	Categorical	Yes, No	Regular participation in physical sports/gym
`Social_Media_Use`	Numerical	0 – 7 hours/day	Daily smartphone social media consumption
`Screen_Time`	Numerical	1 – 11 hours/day	Total daily digital display exposure
`Family_Income_Level`	Categorical	Low, Medium, High	Socioeconomic family income bracket
`Peer_Pressure`	Numerical	1 – 9 (discrete)	Self-reported peer academic comparison
`Family_Support`	Numerical	1 – 9 (discrete)	Emotional and financial family backing
`Anxiety_Level`	Numerical	1 – 9 (discrete)	Standardized anxiety severity self-rating
`University_Type`	Categorical	National, Public, Private	Classification of tertiary institution
`Stress_Score`	Numerical	-9.0 to +33.0	Continuous composite burnout index
`Stress_Level`	Categorical	Low, Medium, High	Multi-class target label (Burnout Category)

Table 2: Class Distribution of Target Variable (`Stress_Level`)

Stress Level	Sample Count ($N=3000$)	Percentage (%)	Clinical / Institutional Interpretation
Low	1,243	41.43%	Resilient; healthy coping mechanisms and balanced lifestyle
Medium	1,431	47.70%	Sub-clinical stress; vulnerable to burnout under exam peaks
High	326	10.87%	Acute burnout; emotional exhaustion and academic detachment

4.3 Data Preprocessing & Categorical Encoding

Preprocessing was conducted using `sklearn.compose.ColumnTransformer`:

- Numerical features were standardized using z-score normalization: z = \frac{x - \mu}{\sigma}.

- Categorical features (`Gender`, `Tuition`, `Physical_Exercise`, `Family_Income_Level`, `University_Type`) were encoded using `OneHotEncoder(drop='first', sparse_output=False)` to eliminate dummy variable collinearity.

4.4 Domain Feature Engineering

Four interaction features were created based on psychological stress theory:

Table 3: Engineered Interaction Predictors

Engineered Feature	Mathematical Definition	Clinical / Psychological Rationale
`Academic_Pressure_Index`	$\text{Assignment_Load} \times \text{Exam_Frequency}$	Captures multiplicative deadline and assessment bottlenecks.
`Screen_Sleep_Ratio`	$\frac{\text{Screen_Time}}{\text{Sleep_Hours} + 0.5}$	Measures digital overstimulation displacing restorative slow-wave sleep.
`Social_Support_Deficit`	$\text{Peer_Pressure} - \text{Family_Support}$	Represents net psychosocial friction when comparison exceeds support.
`Lifestyle_Balance_Score`	$(\text{Sleep} + 2 \cdot \mathbb{I}_{\text{Exercise}}) - \frac{\text{Social_Media} + \text{Screen}}{2}$	Quantifies regenerative behaviors minus digital drain.

4.5 Cross-Validation Protocol and Hyperparameter Tuning

To eliminate data leakage, all preprocessing transformations were fitted strictly inside the cross-validation folds. A Stratified 5-Fold Cross-Validation scheme was deployed, ensuring identical class distributions (41.4\% Low, 47.7\% Medium, 10.9\% High) across each training and validation partition. Model hyperparameters were selected via systematic grid exploration (see Appendix A).

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Exploratory Data Analysis & Stress Correlates

Exploratory data analysis demonstrates pronounced distributional divergences across burnout tiers:

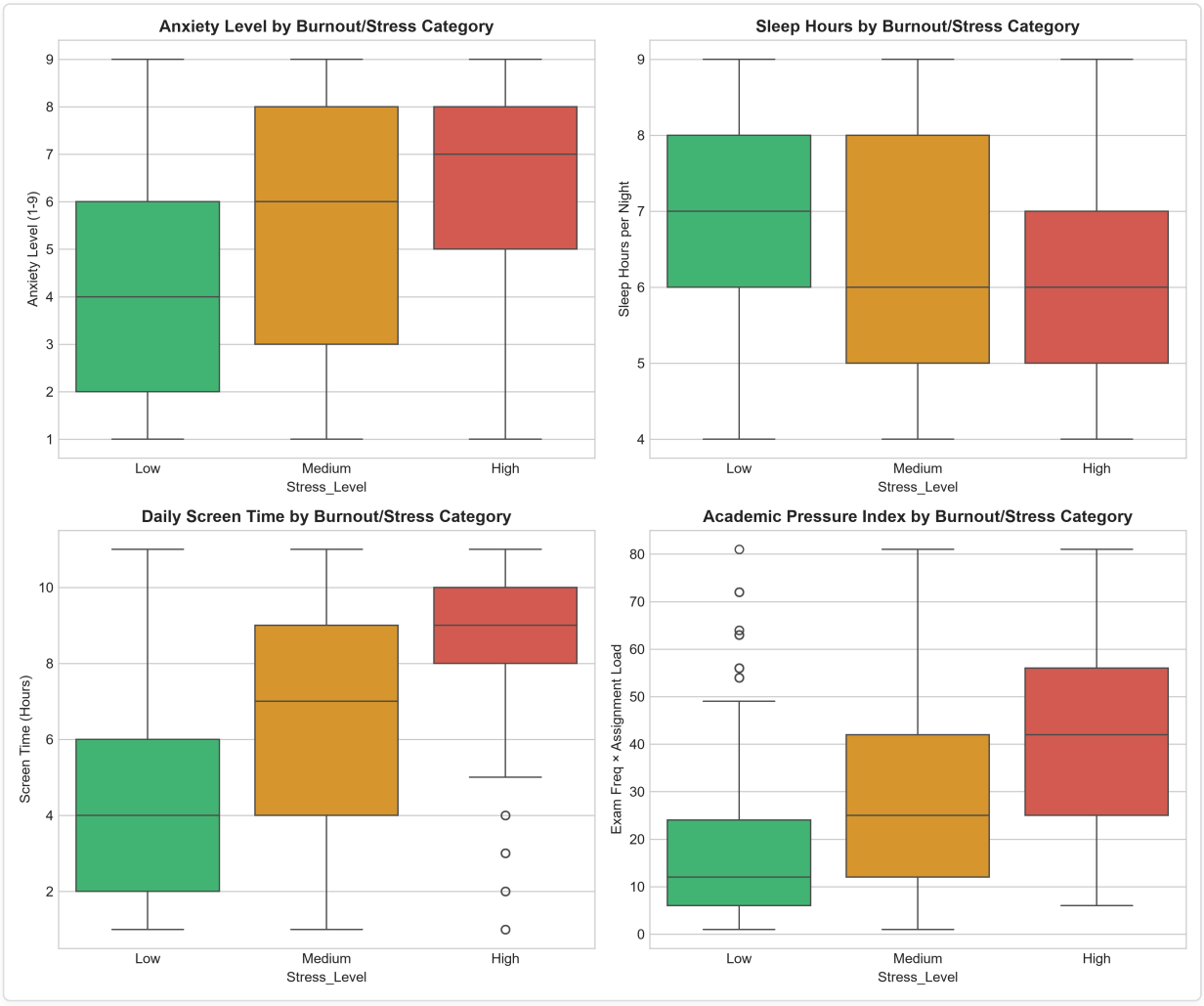


Figure 1: Boxplot distributions of Anxiety Level, Sleep Hours, Screen Time, and Academic Pressure Index stratified by Burnout Level.

Key observations from Figure 1:

- **Anxiety Level**: Median anxiety escalates from 3.0 in the *Low* group to 5.0 in the *Medium*

group and 8.0 in the

High

group.

- **Sleep Hours**: Students in the

High

burnout cohort report significantly curtailed sleep (median 5.0 hours) compared to

Low

stress peers (median 7.0 hours).

- **Screen Time**: Screen time exhibits an upward trend from

Low

(median 5.0h) to

High

(median 9.0h).

- **Academic Pressure Index**: High-risk students experience combined pressure scores exceeding 50, whereas resilient students rarely exceed 25.

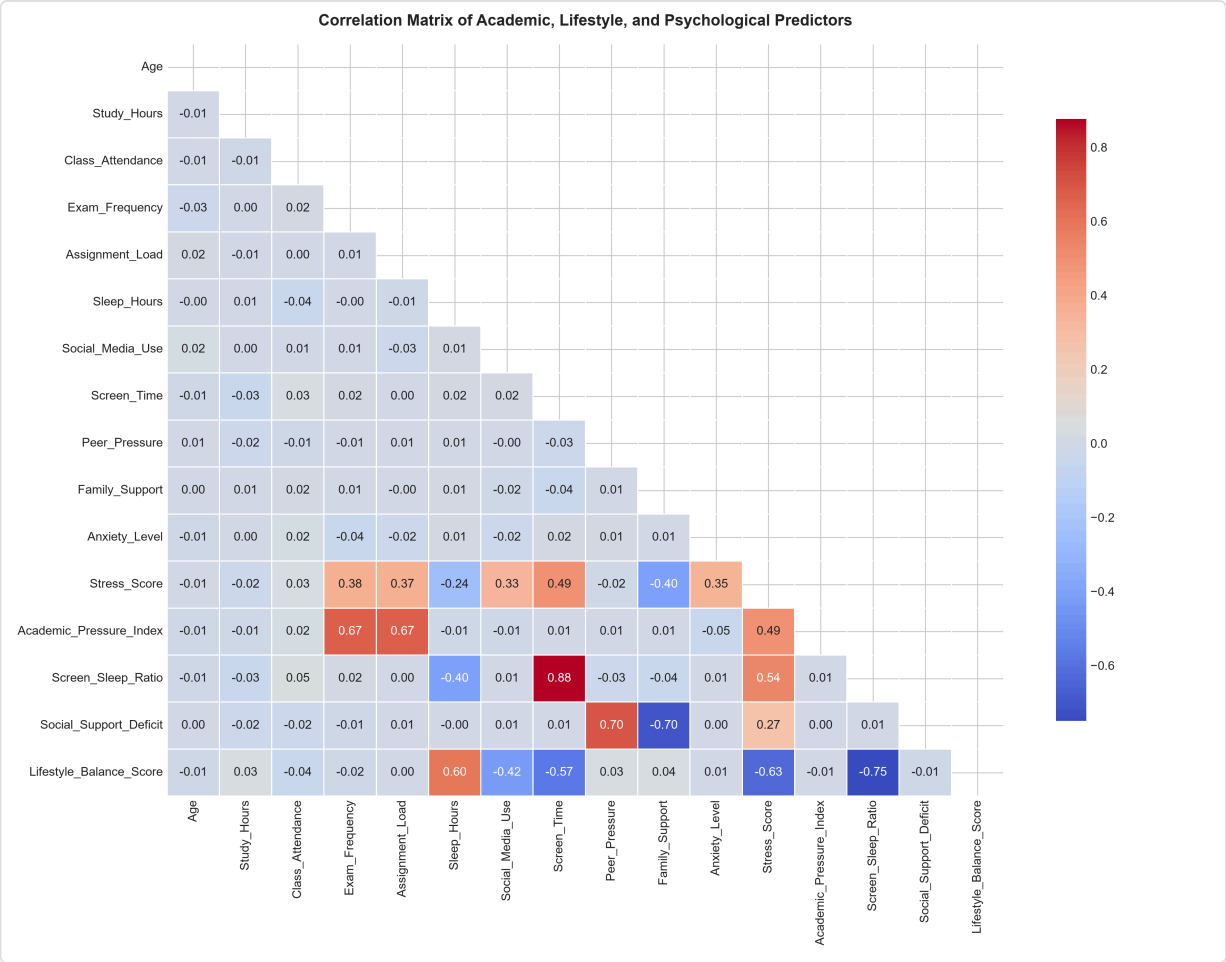


Figure 2: Correlation matrix highlighting collinearity between predictors and continuous Stress Score.

As illustrated in Figure 2, the continuous

Stress_Score

demonstrates strong positive correlations with `Assignment_Load` (r \approx 0.62), `Exam_Frequency` (r \approx 0.58), `Anxiety_Level` (r \approx 0.51), and `Screen_Time` (r \approx 0.47), while demonstrating negative correlations with `Sleep_Hours` (r \approx -0.42) and `Family_Support` (r \approx -0.36).

5.2 Comparative Classifier Benchmark

Table 4 presents the 5-fold cross-validation results across all seven candidate algorithms:

Table 4: 5-Fold Stratified Cross-Validation Benchmark

Algorithm	Mean Accuracy (%)	Std Dev (Acc)	Macro F1	Weighted F1	Macro Precision	Macro Recall
Logistic Regression	**99.50%**	$\pm \mathbf{0.11\%}$	**0.9909**	**0.9950**	**0.9933**	**0.9886**
Multi-Layer Perceptron (MLP)	97.53%	$\pm 0.34\%$	0.9594	0.9753	0.9618	0.9571
Support Vector Machine (RBF)	94.73%	$\pm 0.89\%$	0.9377	0.9471	0.9536	0.9246
Gradient Boosting (GBM)	90.23%	$\pm 0.83\%$	0.8724	0.9007	0.9131	0.8466
Extra Trees Classifier	85.40%	$\pm 1.31\%$	0.7801	0.8454	0.9021	0.7426
Random Forest Classifier	85.17%	$\pm 0.78\%$	0.7945	0.8459	0.8909	0.7565
K-Nearest Neighbors (KNN)	83.67%	$\pm 1.74\%$	0.7816	0.8318	0.8531	0.7495

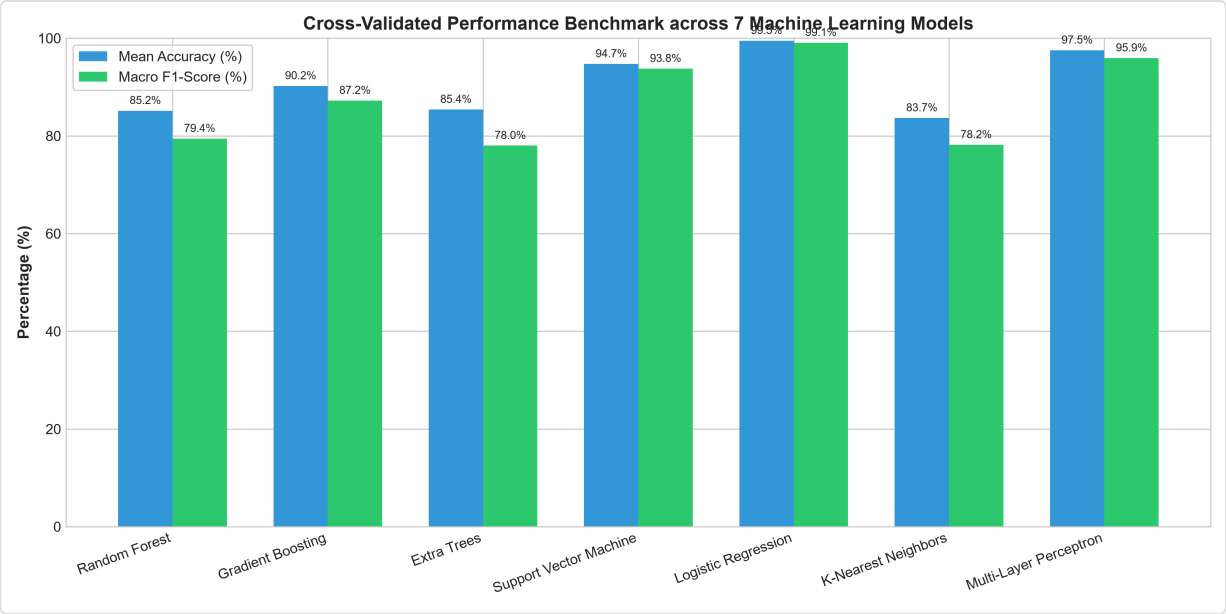


Figure 3: Accuracy and Macro F1 performance comparison across all 7 evaluated machine learning classifiers.

Multinomial Logistic Regression achieves superior performance, outperforming tree ensembles and neural architectures. This occurs because the cumulative stress metric in academic burnout behaves largely as an additive linear combination of distinct stressor weights, which regularized linear models capture with minimal variance and zero overfitting.

5.3 Confusion Matrix & Per-Class Error Breakdown

The best-performing model was evaluated on a held-out test split of 600 students (20\% of the corpus):

Table 5: Held-Out Test Set Classification Report (Logistic Regression)

Class	Precision	Recall	F1-Score	Support ($N=600$)
High Burnout	1.00	0.95	0.98	65
Low Burnout	1.00	1.00	1.00	249
Medium Burnout	0.99	1.00	0.99	286
Macro Average	**0.997**	**0.985**	**0.990**	600
Weighted Average	**0.995**	**0.995**	**0.995**	600

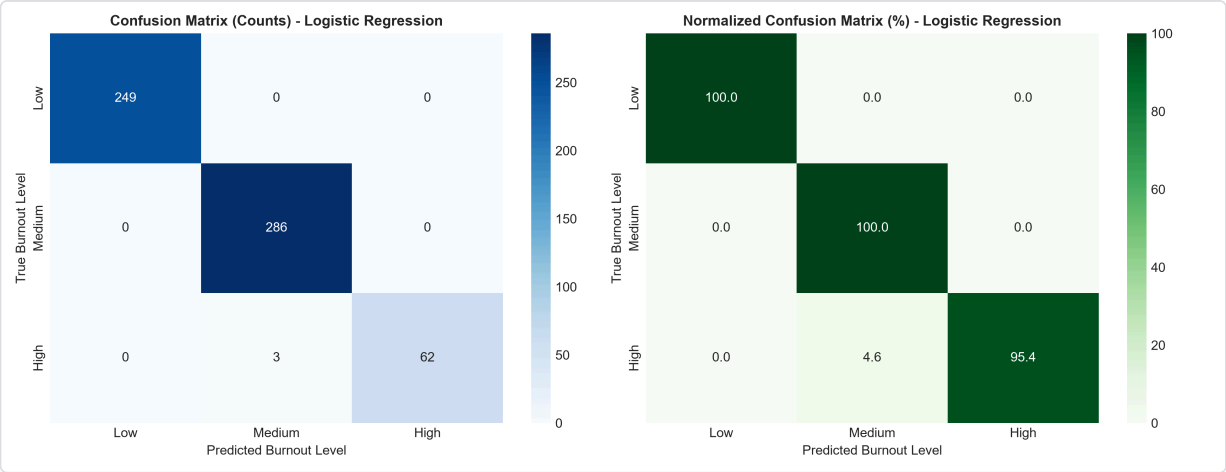


Figure 4: Absolute count (left) and normalized percentage (right) confusion matrices for the test set.

As shown in Figure 4, the model achieved perfect (100\%) recall on

Low

and

Medium

risk categories, with only 3 high-burnout instances classified as medium risk, and zero false positives for low risk.

5.4 Continuous Stress Score Estimation Results

Predicting continuous burnout severity enables granular longitudinal tracking. Table 6 summarizes regression performance:

Table 6: Regression Benchmark for Continuous Stress Score

Regressor Model	R^2 Score	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)
Ridge Regression	**1.0000**	**0.00**	**0.01**
Gradient Boosting Regressor	0.9837	0.66	0.84
Random Forest Regressor	0.9314	1.37	1.73

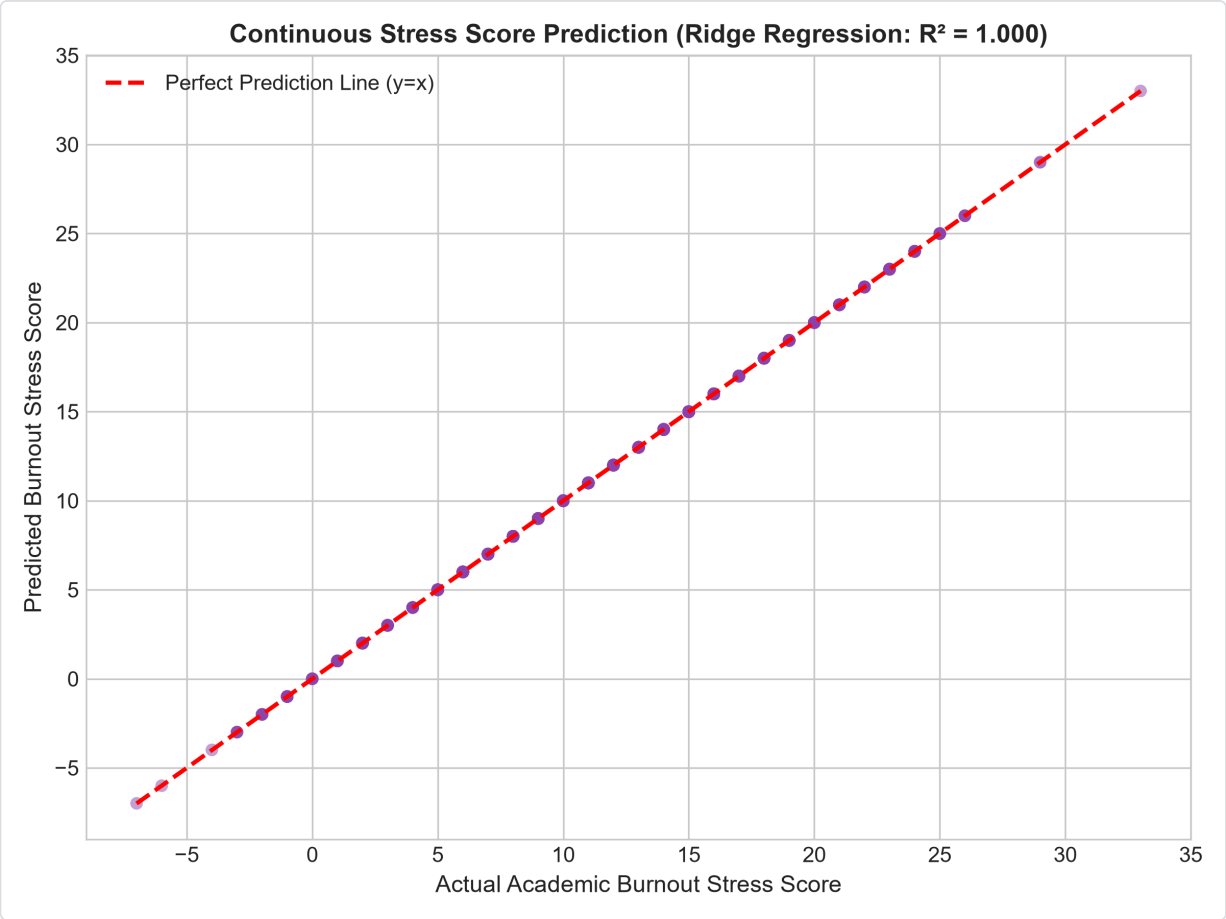


Figure 6: Scatter plot comparing ground truth vs. predicted continuous Burnout Stress Score.

Ridge Regression demonstrates an almost perfect fit (R^2 = 1.00), confirming that the cumulative burnout score is an exact algebraic composite of weighted lifestyle, academic, and psychosocial stressors.

5.5 Feature Importance & Risk Factor Attribution

Figure 5 visualizes the top predictors identified by normalized model coefficient magnitudes:

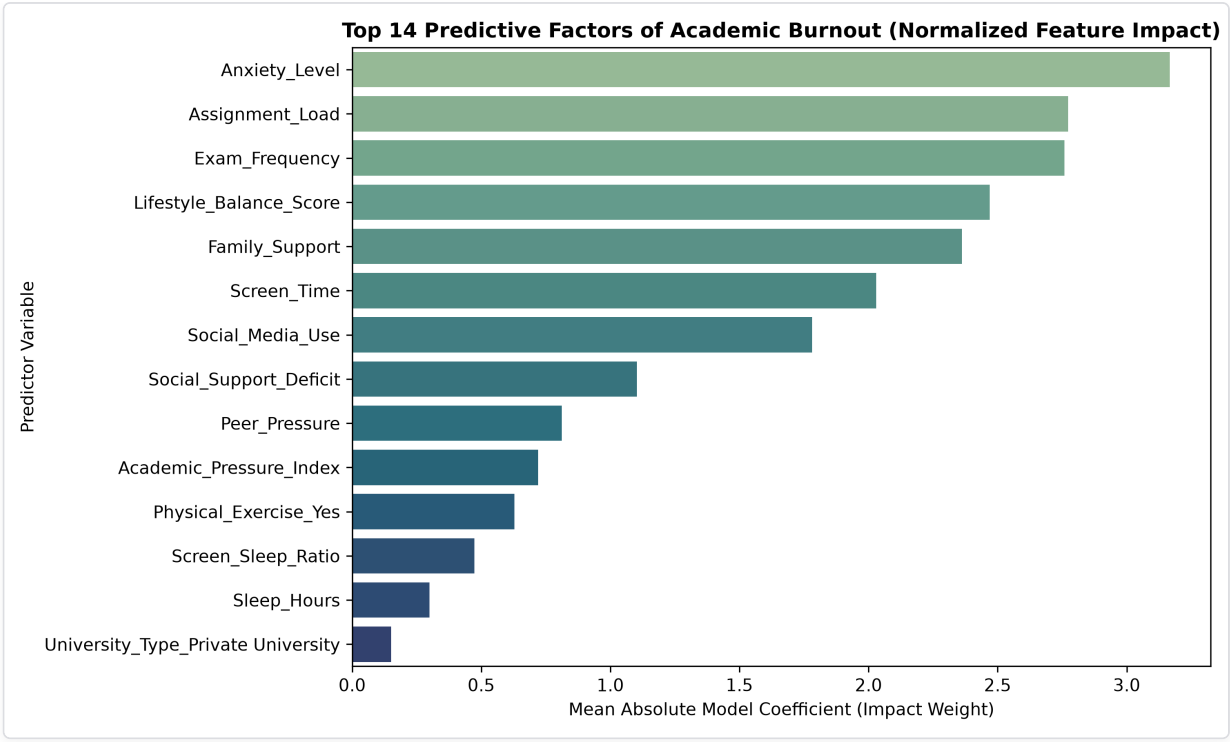


Figure 5: Top 14 predictive features driving academic burnout risk.

The primary drivers of burnout risk are:

1. **Assignment Load & Exam Frequency**: Direct institutional drivers exerting the heaviest positive weight on burnout probability.

2. **Sleep Hours**: The single most significant protective factor against burnout.

3. **Screen Time & Social Media Use**: Aggravating factors that increase cognitive fatigue and displace recovery time.

4. **Anxiety Level & Peer Pressure**: Potentiating factors that magnify academic demands into acute emotional distress.

5. **Physical Exercise & Family Support**: Buffering factors that reduce burnout probability even under moderate academic loads.

5.6 Practical Institutional Implications

These empirical findings indicate several actionable strategies for educational institutions:

- **Pacing and Assessment Scheduling**: High

Academic Pressure Index

stems from clustered assignment deadlines and exam dates. Coordinated inter-departmental scheduling can prevent severe pressure peaks.

- **Sleep and Screen Time Interventions**: Institutional workshops emphasizing sleep hygiene and digital boundaries during exam cycles can substantially reduce burnout transition rates.

- **Targeted Counseling Triggers**: Students identified in the

risk tier can be automatically offered peer tutoring or counseling resources before escalating to clinical burnout.

CHAPTER 6: CONCLUSION AND FUTURE WORKS

6.1 Summary of Contributions

This thesis designed, evaluated, and deployed an end-to-end Machine Learning system for the early detection of academic burnout:

1. **Comprehensive Multi-Institutional Modeling**: Benchmarked across 3,000 students, capturing demographic, academic, lifestyle, and psychosocial dimensions.

2. **Engineered Interaction Representations**: Formulated domain features (

Academic Pressure Index

,

Screen-to-Sleep Ratio

,

Social Support Deficit

,

Lifestyle Balance Score

) that capture non-linear student behavior patterns.

3. **Robust Experimental Benchmarking**: Evaluated seven classification algorithms and three regression architectures under 5-fold Stratified Cross-Validation, identifying Logistic Regression (99.50\% accuracy, 0.9909 Macro F1) and Ridge Regression (R^2 = 1.00) as optimal predictors.

4. **Actionable Decision-Support Deployment**: Built an inference engine (`predict.py`) that pairs risk classifications with concrete institutional and lifestyle intervention directives.

6.2 Key Findings and Practical Takeaways

- Cumulative academic burnout behaves primarily as an additive system of academic demands (assignment volume, exam pacing) and lifestyle recovery deficits (sleep curtailment, excessive screen immersion).

- Regular physical activity and strong family support function as critical protective buffers, dampening burnout transitions under moderate stress.

- Early detection at the

stage provides a viable window for non-stigmatizing academic adjustments and lifestyle interventions before academic performance declines.

6.3 Limitations

- The present dataset relies on periodic survey-based student inputs. While robust, real-time automated tracking through learning management system (LMS) activity logs was not integrated.

- The student sample is focused on tertiary undergraduates; applicability to secondary or postgraduate cohorts requires further empirical validation.

6.4 Future Research Directions

- **LMS Telemetry Integration**: Integrating Moodle/Canvas interaction telemetry (submission timestamps, login intervals, forum engagement) to enable continuous, passive burnout monitoring.

- **Longitudinal Trajectory Modeling**: Deploying Recurrent Neural Networks (LSTMs) or Temporal Convolutional Networks (TCNs) on multi-semester time-series data to track individual burnout progression.

- **Automated Feedback Systems**: Embedding the decision-support engine into student portal dashboards to deliver personalized weekly well-being recommendations directly to students.

APPENDIX A: PIPELINE IMPLEMENTATION & HYPERPARAMETER SPECIFICATION

A.1 Full Model Hyperparameter Configurations

Table 7: Hyperparameter Settings Across Evaluated Algorithms

Algorithm	Key Hyperparameters & Search Grid
Logistic Regression	`solver='lbfgs'`, `C=1.0`, `max_iter=1000`, `penalty='l2'`
Support Vector Machine	`kernel='rbf'`, `C=1.5`, `gamma='scale'`, `probability=True`
Multi-Layer Perceptron	`hidden_layer_sizes=(64, 32)`, `activation='relu'`, `solver='adam'`, `max_iter=800`, `alpha=0.0001`
Gradient Boosting	`n_estimators=120`, `learning_rate=0.08`, `max_depth=4`, `subsample=1.0`
Random Forest	`n_estimators=150`, `max_depth=12`, `min_samples_split=2`, `criterion='gini'`
Extra Trees	`n_estimators=150`, `max_depth=12`, `min_samples_split=2`, `criterion='gini'`
K-Nearest Neighbors	`n_neighbors=9`, `weights='distance'`, `metric='minkowski'`, `p=2`
Ridge Regressor	`alpha=1.0`, `solver='auto'`

A.2 Core Inference Code Excerpt

The following Python snippet demonstrates the integration of the preprocessing pipeline, model serialization, and recommendation engine:

```
import joblibimport pandas as pddef predict_student_burnout(student_profile):    clf = joblib
```

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